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 Studierichting: Informatica
 Oefeningenreeks 5I nr. 2.6

$$L = \int_{\pi/4}^{\pi/3} \sqrt{1 + (f'(x))^2} dx$$

$$f'(x) = \left(-\frac{1}{4} \sin(x) + \ln(\sec(x) + \tan(x)) \right)'$$

$$f'(x) = -\frac{1}{4} \cos(x) + \frac{1}{\sec(x) + \tan(x)} \cdot (\sec(x) \tan(x) + \sec^2(x))$$

$$f'(x) = -\frac{1}{4} \cos(x) + \frac{\sec(x)(\tan(x) + \sec(x))}{\sec(x) + \tan(x)}$$

$$f'(x) = -\frac{1}{4} \cos(x) + \sec(x)$$

$$(f'(x))^2 = \left(-\frac{1}{4} \cos(x) + \sec(x) \right)^2$$

$$(f'(x))^2 = \frac{1}{16} \cos^2(x) - 2 \left(\frac{1}{4} \cos(x) \right) \sec(x) + \sec^2(x)$$

$$(f'(x))^2 = \frac{1}{16} \cos^2(x) - \frac{1}{2} + \sec^2(x)$$

$$1 + (f'(x))^2 = 1 + \frac{1}{16} \cos^2(x) - \frac{1}{2} + \sec^2(x)$$

$$1 + (f'(x))^2 = \frac{1}{16} \cos^2(x) + \frac{1}{2} + \sec^2(x)$$

$$1 + (f'(x))^2 = \left(\frac{1}{4} \cos(x) + \sec(x) \right)^2$$

$$L = \int_{\pi/4}^{\pi/3} \sqrt{\left(\frac{1}{4} \cos(x) + \sec(x) \right)^2} dx$$

$$L = \int_{\pi/4}^{\pi/3} \left(\frac{1}{4} \cos(x) + \sec(x) \right) dx$$

$$L = \left[\frac{1}{4} \sin(x) + \ln |\sec(x) + \tan(x)| \right]_{\pi/4}^{\pi/3}$$

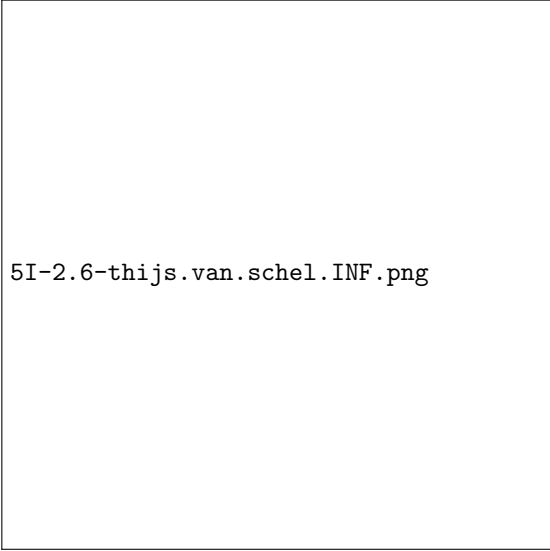
$$L = \left(\frac{1}{4} \sin\left(\frac{\pi}{3}\right) + \ln \left| \sec\left(\frac{\pi}{3}\right) + \tan\left(\frac{\pi}{3}\right) \right| \right) - \left(\frac{1}{4} \sin\left(\frac{\pi}{4}\right) + \ln \left| \sec\left(\frac{\pi}{4}\right) + \tan\left(\frac{\pi}{4}\right) \right| \right)$$

$$L = \left(\frac{1}{4} \left(\frac{\sqrt{3}}{2} \right) + \ln |2 + \sqrt{3}| \right) - \left(\frac{1}{4} \left(\frac{\sqrt{2}}{2} \right) + \ln |\sqrt{2} + 1| \right)$$

$$L = \frac{\sqrt{3}}{8} + \ln(2 + \sqrt{3}) - \frac{\sqrt{2}}{8} - \ln(\sqrt{2} + 1)$$

$$L = \frac{\sqrt{3} - \sqrt{2}}{8} + \ln \left(\frac{2 + \sqrt{3}}{\sqrt{2} + 1} \right)$$

tekening:



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References